

Data Warehousing/ Visualization:
Sales Forecasting using Power BI

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Data Warehousing Visualization

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Introduction.

This report summarizes the analysis of mobile phone sales using Power BI. The dataset contains transactional records with product, customer, and behavioral attributes, enabling time-series forecasting, customer segmentation, and product benchmarking. The objective is to transform raw sales data into actionable insights by identifying sales trends, customer behavior patterns, brand and product performance, and geographical and payment dynamics. A predictive what-if simulation is also included to estimate the revenue impact of pricing and demand changes.

1. Data Model Description and Calculated Columns

The data model was designed following a star schema, with FactSales at the center and supporting dimension tables for Date, Product, Customer, Location, and Payment. This structure ensures efficient filtering, better performance, and scalability for future extensions such as inventory or returns. The Date table was marked as the official date table to enable time intelligence functions (YTD, YoY, seasonality).

To enhance analytical flexibility, several calculated columns were created:

- YearMonth = FORMAT([Date], "YYYY-MM") → supports monthly aggregation for trends and forecasts.
- Quarter = "Q" & FORMAT([Date], "Q") → used in seasonal analysis.
- Age Segment = SWITCH logic to bucket CustomerAge into interpretable ranges (≤17, 18–25, 26–35, 36–45, 46–60, 60+).
- Avg Price per Unit = DIVIDE([TotalRevenue],[UnitsSold]) → provides normalized comparison across products.
- Payment Method Category (optional grouping) → simplifies analysis by consolidating similar payment types.

These calculated columns complement the core KPIs (Total Revenue, Total Units, and Average Price), providing the foundation for segmentation, product ranking, and predictive scenario simulations. Together, the star schema and calculated columns create a flexible model that supports advanced insights while maintaining a simple and efficient structure.

2. Explanation of DAX logic

The analytical foundation of the dashboards relies on a structured set of DAX measures grouped into categories to ensure consistency and reusability.

Core KPIs

- Total Revenue = $\text{SUM}(\text{'Sales' [TotalRevenue]})$
- Total Units = $\text{SUM}(\text{'Sales' [UnitsSold]})$
- Average Price = $\text{DIVIDE}([\text{Total Revenue}], [\text{Total Units}], 0)$

These baseline metrics are referenced by all other measures, so totals reconcile across pages.

Time Intelligence

- Revenue YTD = $\text{TOTALYTD}([\text{Total Revenue}], \text{'Date' [Date]})$
- Revenue YoY % = $(\text{Revenue this year} - \text{Revenue last year}) \div \text{Revenue last year}$
- Units Seasonality Ratio = $\text{Units} \div \text{12-month moving average}$

These calculations capture seasonality and growth trends, supporting forecasting and comparisons over time.

Segmentation & Shares

- Age and gender shares, Payment Method %, and Units/Revenue % by segment
normalize values against the appropriate denominator. This provides comparable percentages and avoids misleading totals.

Ranking & Top-N

- Rank Brand by Units and Rank Model by Revenue (RANKX logic) ensures leaderboards remain consistent.
- A dynamic TopN parameter allows users to choose how many products to display without editing DAX.

Scenario Simulation (What-If)

- Parameters for Price Change % and Unit Change % feed into measures such as Sim Revenue = Revenue $\times$ (1 + Price %) $\times$ (1 + Unit %).
- Delta measures quantify the expected lift or decline in revenue compared to baseline.

This layered DAX logic enables every visual to be tied to standardized metrics, ensuring coherence across dashboards and making the model easier to maintain.

3. Forecasting Approach and Assumptions

To project future sales performance, we applied Power BI's built-in forecasting tool on monthly aggregated data for both Total Revenue and Units Sold. This method uses exponential smoothing to extend existing time series, producing a forecast curve with a confidence interval.

Approach:

- Data was aggregated by YearMonth using the Date table.
- Forecast length was set to 6–12 months to capture short-term trends.
- Seasonality was detected automatically, with manual checks for recurring monthly peaks.

Assumptions:

- Past trends are a reliable indicator of short-term future sales.

- Seasonal cycles (e.g., demand peaks in specific months) remain consistent over the forecast horizon.
- No structural market shocks or abrupt product portfolio changes occur in the forecast period.

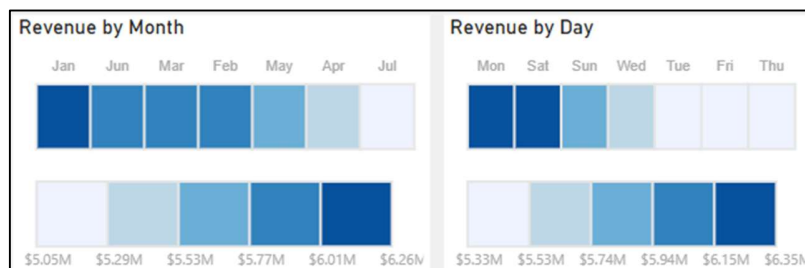
This forecasting approach provides decision-makers with a data-driven forward view of expected performance, while acknowledging that results depend on the stability of underlying trends and external conditions.

4. Key Insights and Visualizations

This forecasting approach provides decision-makers with a data-driven forward view of expected performance, while acknowledging that results depend on the stability of underlying trends and external conditions.

a. Sales trend and forecasting

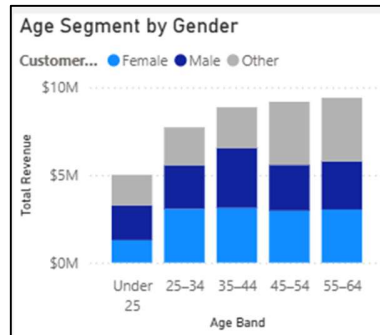
The analysis shows clear seasonal patterns in mobile sales. January emerges as the strongest month in terms of revenue contribution, while Mondays consistently drive the highest weekly sales. This highlights key seasonal and weekly patterns that can guide inventory planning and marketing strategies.



b. Customer Segmentation

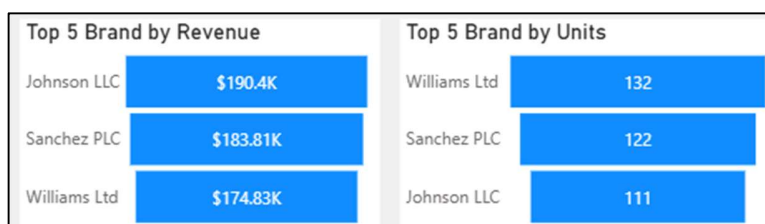
Revenue contribution is balanced across genders, with Female and Other segments generating slightly higher revenue than Male customers. However, Male customers purchase

a higher number of units on average. The 25–34 and 35–44 age groups stand out as the most valuable segments, driving the largest revenue share across all genders.

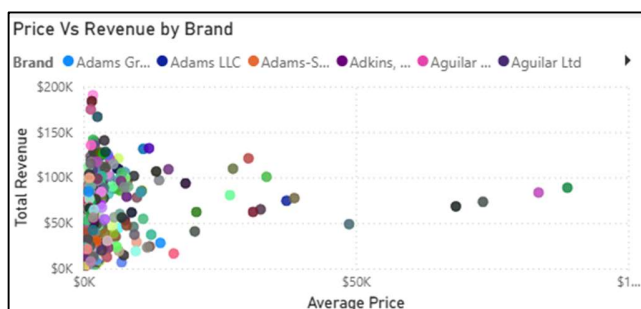


c. Products & Pricing

Johnson LLC leads in total revenue (\$190.9K), while Williams Ltd dominates unit sales (132), showing that high sales volume does not always translate into the highest revenue.



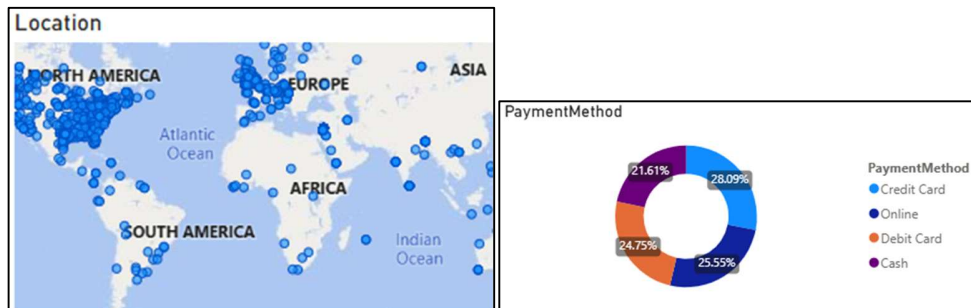
The scatter plot of Price vs Revenue suggests that brands with mid-range pricing achieve the best revenue performance, while higher-priced models generate lower sales volumes.



d. Geography and payment methods trends

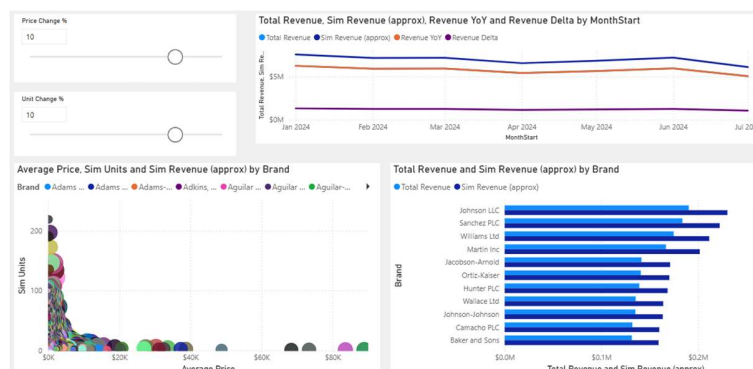
In terms of behavior, credit cards and online payments are the dominant methods, highlighting customer preference for digital and convenient transactions. Geographically,

revenue distribution shows variation by location, but no single region overwhelmingly dominates, suggesting opportunities for localized marketing strategies.



e. What-if Simulator

A sensitivity analysis simulating $\pm 10\%$ changes in prices and units revealed that variations in sales volume have a stronger proportional impact on revenue compared to equivalent price adjustments, highlighting that volume-driven strategies may be more effective for boosting overall revenue than relying solely on aggressive pricing changes.



Conclusion.

The analysis demonstrates that customer segmentation, product mix, and predictive simulations provide valuable insights into revenue drivers, emphasizing the importance of volume growth strategies and balanced pricing to sustain long-term performance.

References

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